

121. Let $f : \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ be a function satisfying $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$ for all $x, y, f(y) \neq 0$. If $f'(1) = 2024$, then :

माना एक फलन $f : \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ सभी $x, y, f(y) \neq 0$ के लिए $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$ को संतुष्ट करता है। यदि $f'(1) = 2024$ है, तो

:

(A) $xf'(x) + f(x) = 2024$

(B) $xf'(x) - 2024f(x) = 0$

(C) $xf'(x) + 2024f(x) = 0$

(D) $xf'(x) - 2023f(x) = 0$

Ans. B

Sol. $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$

$f(1) = 2024$

$f(1) = 1$

Partially differentiating w.r.t. x

$$f'\left(\frac{x}{y}\right) \cdot \frac{1}{y} = \frac{1}{f(y)} f'(x)$$

$y \rightarrow x$

$$f'(1) \cdot \frac{1}{x} = \frac{f'(x)}{f(x)}$$

$$2024f(x) = xf'(x) \Rightarrow xf'(x) - 2024f(x) = 0$$

122. Consider the function $f : (0, \infty) \rightarrow \mathbb{R}$ defined by $f(x) = e^{-|\log_e x|}$. If m and n be respectively the number of points at which f is not continuous and f is not differentiable, then $m + n$ is :

$f(x) = e^{-|\log_e x|}$ द्वारा परिभाषित फलन $f : (0, \infty) \rightarrow \mathbb{R}$ का विचार कीजिए। यदि उन बिंदुओं की संख्या, जहाँ f संतत नहीं है तथा अवकलनीय नहीं हैं क्रमशः m तथा n है, तो $m + n$ बराबर है :

(A) 2

(B) 3

(C) 0

(D) 1

Ans. D

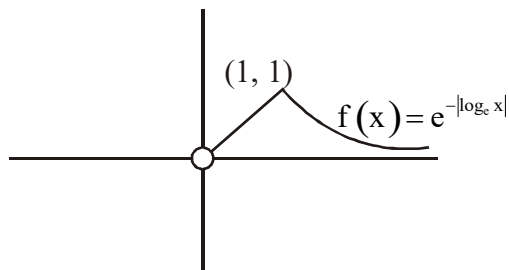
Sol. $f(x) = e^{-|\ln x|}$

$f : (0, \infty) \rightarrow \mathbb{R}$

$f(x) = e^{-|\log_e x|}$

$$f(x) = \frac{1}{e^{|\ln x|}} = \begin{cases} \frac{1}{e^{-\ln x}}, & 0 < x < 1 \\ \frac{1}{e^{\ln x}}, & x \geq 1 \end{cases}$$

$$= \begin{cases} x, & 0 < x < 1 \\ \frac{1}{x}, & x \geq 1 \end{cases}$$



$f(x)$ is continuous in $(0, \infty) \Rightarrow m = 0$

from graph, $f(x)$ is not differentiable at 1 point at $x = 1$

$\Rightarrow n = 1$

therefore $m + n = 1$

123. Let $f(x) = |2x^2 + 5|x| - 3|$, $x \in \mathbb{R}$. If m and n denote the number of points where f is not continuous and not differentiable respectively, then $m + n$ is equal to :

माना $f(x) = |2x^2 + 5|x| - 3|$, $x \in \mathbb{R}$ हैं। यदि m तथा n क्रमशः उन बिंदुओं, जहाँ पर f संतत नहीं है तथा अवकलनीय नहीं है की संख्या हैं, तो $m + n$ बराबर है :

(A) 5

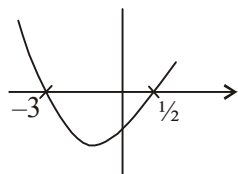
(B) 0

(C) 3

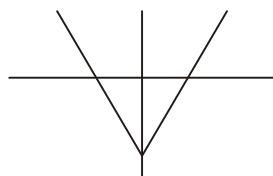
(D) 2

Ans. C

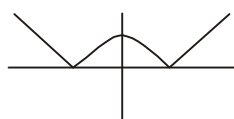
Sol. $y = 2x^2 + 5|x| - 3$



$$y = 2x^2 + 5|x| - 3$$



$$y = |2x^2 + 5|x| - 3|$$



$$m = 0, n = 3$$

$$\Rightarrow m + n = 3$$

124. Let m and n be the number of points at which the function $f(x) = \max \{x, x^3, x^5, \dots, x^{21}\}, x \in \mathbb{R}$, is not differentiable and not continuous, respectively. Then $m + n$ is equal to _____.

माना उन बिन्दुओं, जहाँ पर फलन $f(x) = \max \{x, x^3, x^5, \dots, x^{21}\}, x \in \mathbb{R}$, अवकलनीय नहीं है और संतत नहीं है, की संख्या क्रमशः m और n है। तो $m + n$ बराबर है _____.

Ans. 3

Sol. $f(x) = \max \{x, x^3, x^5, \dots, x^{21}\}$

$$f(x) = \begin{cases} x & ; \quad x < -1 \\ x^{21} & ; \quad -1 \leq x < 0 \\ x & ; \quad 0 \leq x < 1 \\ x^{21} & ; \quad x \geq 1 \end{cases}$$

$f(x)$ is continuous everywhere
 $n = 0$

$$f'(x) = \begin{cases} 1 & ; \quad x < -1 \\ 21x^{20} & ; \quad -1 \leq x < 0 \\ 1 & ; \quad 0 < x < 1 \\ 21x^{20} & ; \quad x \geq 1 \end{cases}$$

$f(x)$ is non-differentiable at $x = -1, 0, 1$

$$\therefore m = 3$$

$$m + n = 3$$

125. The number of points of discontinuity of the function $f(x) = \left\lfloor \frac{x^2}{2} \right\rfloor - \left\lfloor \sqrt{x} \right\rfloor, x \in [0, 4]$, where $\lfloor \bullet \rfloor$ denotes the greatest integer function, is _____.

फलन $f(x) = \left\lfloor \frac{x^2}{2} \right\rfloor - \left\lfloor \sqrt{x} \right\rfloor, x \in [0, 4]$, जहाँ where $\lfloor \bullet \rfloor$ महत्तम पूर्णांक फलन है, के असांतत्य के बिन्दुओं की संख्या है _____.

Ans. 8

Sol. $x = 0, 1, \sqrt{2}, \sqrt{4}, \sqrt{6}, \sqrt{8}, \sqrt{10}, \sqrt{12}, \sqrt{14}, 4$

$$0 \leq x^2 \leq 16$$

$$0 \leq \frac{x^2}{2} \leq 8$$

$$x^2 = 0, 2, 4, 6, 8, 10, 12, 14, 16$$

$$x = 0, \sqrt{2}, 2, \sqrt{6}, 2\sqrt{2}, \sqrt{10}, 2\sqrt{3}, \sqrt{14}, 4$$

$$\frac{x^2}{2} = 0, 1, 2, 3, 4, 5, 6, 7, 8 \quad \rightarrow \quad 8 \text{ points}$$

$$0 \leq \sqrt{x} \leq 2$$

$$\sqrt{x} = 0, 1, 2 \quad \rightarrow \quad 2 \text{ points}$$

$$x = 0, 1, 4$$

$$f(4) = 8 - 2 = 6$$

$$f(4^-) = 7 - 1 = 6$$

$$\text{Total points} = 8 + 2 - 2 = 8 \text{ points}$$

126. If the function $f(x) = \frac{\tan(\tan x) - \sin(\sin x)}{\tan x - \sin x}$ is continuous at $x = 0$, then $f(0)$ is equal to _____.

यदि फलन $f(x) = \frac{\tan(\tan x) - \sin(\sin x)}{\tan x - \sin x}$, $x = 0$ पर संतत है, तो $f(0)$ बराबर है _____.

Ans. 2

Sol. $f(x) = \lim_{x \rightarrow 0} \frac{\tan(\tan x) - \sin(\sin x)}{\tan x - \sin x}$

by expansion

$$= \lim_{x \rightarrow 0} \frac{\left(\tan x + \frac{(\tan x)^3}{3} + \dots \right) - \left(\sin x - \frac{(\sin x)^3}{3!} \right)}{\left(x + \frac{x^3}{3} - \left(x - \frac{x^3}{3!} \right) \right)}$$

$$= \lim_{x \rightarrow 0} \frac{\left(x + \frac{x^3}{3} \right) + \frac{1}{3} \left(x + \frac{x^3}{3} \right)^3 - \left(x - \frac{x^3}{3!} \right) + \frac{1}{6} \left(x - \frac{x^3}{6} \right)^3}{\frac{x^3}{2}}$$

$$= \lim_{x \rightarrow 0} \frac{x^3 \left[\frac{1}{3} + \frac{1}{3} + \frac{1}{6} + \frac{1}{6} \right]}{\frac{x^3}{2}} = 2$$

127. Let $f(x) = \begin{cases} 3x, & x < 0 \\ \min\{1+x+[x], x+2[x]\}, & 0 \leq x \leq 2 \\ 5, & x > 2, \end{cases}$

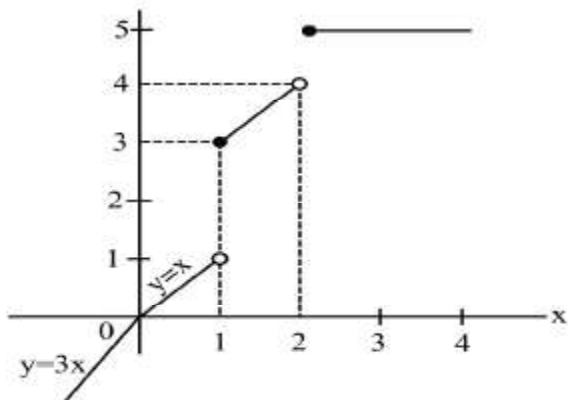
where $[.]$ denotes greatest integer function. If α and β are the number of points, where f is not continuous and is not differentiable, respectively, then $\alpha + \beta$ equals _____.

माना कि $f(x) = \begin{cases} 3x, & x < 0 \\ \min\{1+x+[x], x+2[x]\}, & 0 \leq x \leq 2 \\ 5, & x > 2, \end{cases}$ है,

जहाँ $[.]$ महत्तम पूर्णांक फलन को निर्दिष्ट करता है। यदि α और β क्रमशः उन बिन्दुओं की संख्याएँ हैं, जहाँ f संतत नहीं है और अवकलनीय नहीं है, तो $\alpha + \beta$ बराबर है _____.

Ans. 5

Sol. $f(x) = \begin{cases} 3x & ; & x < 0 \\ x & ; & 0 \leq x < 1 \\ x+2 & ; & 1 \leq x < 2 \\ 5 & ; & x > 2 \end{cases}$



Not continuous at $x \in \{1, 2\} \Rightarrow \alpha = 2$

Not differentiable at $x \in \{0, 1, 2\} \Rightarrow \beta = 3$

$$\alpha + \beta = 5$$

128. Let $[t]$ denote the greatest integer less than or equal to t . Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \left\lfloor \frac{x}{2} + 3 \right\rfloor - \lfloor \sqrt{x} \rfloor. \text{ Let } S \text{ be the set of all points in the interval } [0, 8] \text{ at which } f \text{ is not continuous. Then } \sum_{a \in S} a$$

is equal to _____.

माना $[t]$, t से कम या t के बराबर महत्तम पूर्णांक को दर्शाता है। माना फलन $f : [0, \infty) \rightarrow \mathbb{R}$, $f(x) = \left[\frac{x}{2} + 3 \right] - [\sqrt{x}]$ द्वारा परिभाषित है। माना S , अन्तराल $[0, 8]$ में उन सभी बिन्दुओं का समुच्चय है, जहाँ f संतत नहीं है। तो $\sum_{a \in S} a$ बराबर है _____

Ans. 17

Sol. $f(x) = \left[\frac{x}{2} + 3 \right] - [\sqrt{x}]$

$$= \left[\frac{x}{2} \right] - [\sqrt{x}] + 3 \quad \because x \in [0, 8], \frac{x}{2} \in [0, 4], \sqrt{x} \in [0, 2\sqrt{2}]$$

Point of discontinuity

When $\frac{x}{2}$ is integer & $\sqrt{x}$ is int.

$$\frac{x}{2} = 0, 1, 2, 3, 4 \quad \sqrt{x} = 0, 1, 4, 9$$

$$x = \underbrace{0}_{\substack{\text{Right} \\ \text{cont}}}, \underbrace{2, 4, 6, 8}_{\text{discont}} \quad x = \underbrace{0}_{\substack{\text{cont}}}, \underbrace{1, 4, 9}_{\text{discont}}$$

$\therefore f(x)$ is discontinuous at $x = 1, 2, 6, 8$

Check continuity at $x = 4$

$$f(4^+) = [2^+] - [\sqrt{4^+}] + 3 = 3$$

$$f(4^-) = [2^-] - [\sqrt{4^-}] + 3 = 1 - 1 + 3 = 3$$

$$f(4) = [2] - [\sqrt{4}] + 3 = 3$$

$\therefore f(x)$ is continuous at $x = 4$

Point of discontinuity = 1, 2, 6, 8

$$\sum_{a \in S} a = 1 + 2 + 6 + 8 = 17$$

129. Let $f : (0, \pi) \rightarrow \mathbb{R}$ be a function given by $f(x) = \begin{cases} \left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0 < x < \frac{\pi}{2} \\ a - 8, & x = \frac{\pi}{2} \\ (1 + |\cot x|)^{\frac{b}{a}|\tan x|}, & \frac{\pi}{2} < x < \pi \end{cases}$ where $a, b \in \mathbb{Z}$. If f is continuous

at $x = \frac{\pi}{2}$, then $a^2 + b^2$ is equal to _____.

माना $f : (0, \pi) \rightarrow \mathbb{R}$ एक फलन दिया गया है जो कि $f(x) = \begin{cases} \left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0 < x < \frac{\pi}{2} \\ a - 8, & x = \frac{\pi}{2} \\ (1 + |\cot x|)^{\frac{b}{a}|\tan x|}, & \frac{\pi}{2} < x < \pi \end{cases}$ है, जहाँ $a, b \in \mathbb{Z}$ है। यदि

$x = \frac{\pi}{2}$ पर संतत है, तो $a^2 + b^2$ is बराबर है _____

Ans. 81

Sol. As $f(x)$ is continuous at x

$$\Rightarrow Lf\left(\frac{\pi}{2}\right) = f\left(\frac{\pi}{2}\right) = Rf\left(\frac{\pi}{2}\right)$$

$$\Rightarrow \lim_{x \rightarrow \frac{\pi}{2}^-} \left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}} = a - 8 = \lim_{x \rightarrow \frac{\pi}{2}^+} (1 + |\cot x|)^{\frac{b}{a}|\tan x|}$$

$$\Rightarrow \left(\frac{8}{7}\right)^0 = a - 8 = e^{\frac{b}{a}}$$

$$\Rightarrow a = 9 \text{ and } b = 0$$

$$\text{Hence } a^2 + b^2 = 81$$

130. Let for a differentiable function $f: (0, \infty) \rightarrow \mathbb{R}$, $f(x) - f(y) \geq \log_e \left(\frac{x}{y} \right) + x - y$, $\forall x, y \in (0, \infty)$. Then $\sum_{n=1}^{20} f' \left(\frac{1}{n^2} \right)$ is equal to _____.

माना एक अवकलनीय फलन $f: (0, \infty) \rightarrow \mathbb{R}$ के लिए, $f(x) - f(y) \geq \log_e \left(\frac{x}{y} \right) + x - y$, $\forall x, y \in (0, \infty)$ है। तो $\sum_{n=1}^{20} f' \left(\frac{1}{n^2} \right)$ बराबर है _____।

Ans. 2890

Sol. $f(x) - f(y) \geq \ln x - \ln y + x - y$

$$\frac{f(x) - f(y)}{x - y} \geq \frac{\ln x - \ln y}{x - y} + 1$$

Let $x > y$

$$\lim_{y \rightarrow x} f'(x^-) \geq \frac{1}{x} + 1 \quad \text{_____ (1)}$$

Let $x < y$

$$\lim_{y \rightarrow x} f'(x^+) \leq \frac{1}{x} + 1 \quad \text{_____ (2)}$$

$$f'(x^-) = f'(x^+)$$

$$f'(x) = \frac{1}{x} + 1$$

$$f' \left(\frac{1}{x^2} \right) = x^2 + 1$$

$$\sum_{x=1}^{20} (x^2 + 1) = \sum_{x=1}^{20} x^2 + 20$$

$$= \frac{20 \times 21 \times 41}{6} + 20$$

$$= 2890$$

Method of Differentiation

131. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such $(\sin x \cos y) (f(2x + 2y) - f(2x - 2y)) = (\cos x \sin y) (f(2x + 2y) + f(2x - 2y))$, for all $x, y \in \mathbb{R}$.

If $f'(0) = \frac{1}{2}$, then the value of $24f'' \left(\frac{5\pi}{3} \right)$ is :

माना एक फलन $f: \mathbb{R} \rightarrow \mathbb{R}$ दो बार अवकलनीय है तथा सभी $x, y \in \mathbb{R}$ के लिए

$(\sin x \cos y) (f(2x+2y) - f(2x-2y)) = (\cos x \sin y) (f(2x+2y) + f(2x-2y))$ है। यदि $f'(0) = \frac{1}{2}$ है, तो $24f''\left(\frac{5\pi}{3}\right)$

का मान है :

(A) 2

(B) -2

(C) 3

(D) -3

Ans. D

Sol. $(\sin x \cos y) (f(2x+2y) - f(2x-2y)) = (\cos x \sin y) (f(2x+2y) + f(2x-2y))$

$$f(2x+2y)(\sin(x-y)) = f(2x-2y) \sin(x+y)$$

$$\frac{f(2x+2y)}{\sin(x+y)} = \frac{f(2x-2y)}{\sin(x-y)}$$

$$\text{Put } 2x+2y = m, 2x-2y = n$$

$$\frac{f(m)}{\sin\left(\frac{m}{2}\right)} = \frac{f(n)}{\sin\left(\frac{n}{2}\right)} = K$$

$$\Rightarrow f(m) = K \sin\left(\frac{m}{2}\right)$$

$$\therefore f(x) = K \sin\left(\frac{x}{2}\right)$$

$$\text{Put } x = 0; \frac{1}{2} = \frac{K}{2} \Rightarrow K = 1$$

$$f'(x) = \frac{1}{2} \cos \frac{x}{2}$$

$$f''(x) = -\frac{1}{4} \sin \frac{x}{2}$$

$$4f''\left(\frac{5\pi}{3}\right) = \left(-\frac{1}{4} \sin\left(\frac{5\pi}{6}\right)\right) 24$$

$$= -\frac{24}{8} = -3$$

132. If $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}$, $x \in \mathbb{R}$, then $\frac{d^2y}{dx^2} + y$ is equal to :

यदि $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}$, $x \in \mathbf{R}$ है, तो $\frac{d^2y}{dx^2} + y$ बराबर है :

- (A) -1 (B) 28 (C) 27 (D) 1

Ans. A

Sol. $C_3 \rightarrow C_3 - C_1$

$$y(x) = \begin{vmatrix} \sin x & \cos x & 1 + \cos x \\ 27 & 28 & 0 \\ 1 & 1 & 0 \end{vmatrix}$$

$$y(x) = -(1 + \cos x)$$

$$\frac{dy}{dx} = \sin x \quad \Rightarrow \quad \frac{d^2y}{dx^2} = \cos x$$

$$\frac{d^2y}{dx^2} + y = -1$$

133. Let $f(x) = x^5 + 2x^3 + 3x + 1$, $x \in \mathbf{R}$, and $g(x)$ be a function such that $g(f(x)) = x$ for all $x \in \mathbf{R}$. the $\frac{g(7)}{g'(7)}$ is equal to :

माना $f(x) = x^5 + 2x^3 + 3x + 1$, $x \in \mathbf{R}$ है, तथा एक फलन $g(x)$ इस प्रकार है कि $g(f(x)) = x \forall x \in \mathbf{R}$ है। तो $\frac{g(7)}{g'(7)}$ बराबर है :

- (A) 42 (B) 7 (C) 14 (D) 1

Ans. C

Sol. $f(x) = x^5 + 2x^3 + 3x + 1$

$$f'(x) = 5x^4 + 6x^2 + 3$$

$$f'(1) = 5 + 6 + 3 = 14$$

$$g(f(x)) = x$$

$$g'(f(x))f'(x) = 1$$

$$\text{for } f(x) = 7$$

$$\Rightarrow x^5 + 2x^3 + 3x + 1 = 7$$

$$\Rightarrow x = 1$$

$$g'(7) f(1) = 1 \Rightarrow g'(7) = \frac{1}{f'(1)} = \frac{1}{14}$$

$$x = 1, f(x) = 7 \Rightarrow g(7) = 1$$

$$\frac{g(7)}{g'(7)} = \frac{1}{1/14} = 14$$

134. If $y(\theta) = \frac{2 \cos \theta + \cos 2\theta}{\cos 3\theta + 4 \cos 2\theta + 5 \cos \theta + 2}$, then at $\theta = \frac{\pi}{2}$, $y'' + y' + y$ is equal to :

यदि $y(\theta) = \frac{2 \cos \theta + \cos 2\theta}{\cos 3\theta + 4 \cos 2\theta + 5 \cos \theta + 2}$ है, तो $\theta = \frac{\pi}{2}$, $y'' + y' + y$ बराबर है :

(A) 2 (B) 1/2 (C) 1 (D) 3/2

Ans. A

Sol. $y = \frac{2 \cos \theta + 2 \cos^2 \theta - 1}{4 \cos^3 \theta - 3 \cos \theta + 8 \cos^2 \theta - 4 + 5 \cos \theta + 2}$

$$y = \frac{(2 \cos^2 \theta + 2 \cos \theta - 1)}{(2 \cos^2 \theta + 2 \cos \theta - 1)(2 \cos \theta + 2)}$$

$$y = \frac{1}{2} \left(\frac{1}{1 + \cos \theta} \right)$$

$$\Rightarrow \theta = \frac{\pi}{2} \quad y = \frac{1}{2}$$

$$y' = \frac{1}{2} \left(\frac{-1}{(1 + \cos \theta)^2} \times (-\sin \theta) \right)$$

$$\Rightarrow \theta = \frac{\pi}{2} \quad y' = \frac{1}{2}$$

$$y'' = \frac{1}{2} \left[\frac{\cos \theta (1 + \cos \theta)^2 - \sin \theta (2)(1 + \cos \theta)(-\sin \theta)}{(1 + \cos \theta)^4} \right]$$

$$\Rightarrow \theta = \frac{\pi}{2} \quad y'' = 1$$

135. Suppose for a differentiable function h , $h(0)=0$, $h(1)=1$ and $h'(0) = h'(1) = 2$. If $g(x) = h(e^x) e^{h(x)}$, then $g'(0)$

is equal to :

मान लीजिए कि अवकलनीय फलन h के लिए, $h(0)=0$, $h(1)=1$ तथा $h'(0) = h'(1) = 2$ हैं। यदि $g(x) = h(e^x) e^{h(x)}$, तो $g'(0)$ बराबर है :

- (A) 3 (B) 8 (C) 4 (D) 5

Ans. C

Sol. $g(x) = h(e^x) \cdot e^{h(x)}$
 $g'(x) = h(e^x) \cdot e^{h(x)} \cdot h'(x) + e^{h(x)} h'(e^x) \cdot e^x$
 $g'(0) = h(1) e^{h(0)} h'(0) + e^{h(0)} h'(1)$
 $= 2 + 2 = 4$

136. Let $f(x) = ax^3 + bx^2 + cx + 41$ be such that $f(1) = 40$, $f'(1) = 2$ and $f''(1) = 4$. Then $a^2 + b^2 + c^2$ is equal to :

माना $f(x) = ax^3 + bx^2 + cx + 41$ इस प्रकार है कि $f(1) = 40$, $f'(1) = 2$ तथा $f''(1) = 4$ है। तो $a^2 + b^2 + c^2$ बराबर है :

- (A) 73 (B) 62 (C) 51 (D) 54

Ans. C

Sol. $f(x) = ax^3 + bx^2 + cx + 41$
 $f'(x) = 3ax^2 + 2bx + c$
 $\Rightarrow f(1) = 3a + 2b + c = 2$ (1)
 $f''(x) = 6ax + 2b$
 $\Rightarrow f''(1) = 6a + 2b = 4$
 $3a + b = 2$... (2)
 $(A) - (2) \Rightarrow b + c = 0$... (3)
 $f(1) = 40$
 $a + b + c + 41 = 40$
 use (3)
 $a + 41 = 40$
 by (2)
 $-3 + b = 2 \Rightarrow b = 5$ & $c = -5$
 $a^2 + b^2 + c^2 = 1 + 25 + 25 = 51$

137. If $\log_e y = 3 \sin^{-1} x$, then $(1 - x^2)y'' - xy'$ at $x = \frac{1}{2}$ is equal to :

यदि $\log_e y = 3 \sin^{-1} x$ है, तो $(1 - x^2)y'' - xy'$ at $x = \frac{1}{2}$ बराबर है :

- (A) $9e^{\frac{\pi}{6}}$ (B) $3e^{\frac{\pi}{2}}$ (C) $3e^{\frac{\pi}{6}}$ (D) $9e^{\frac{\pi}{2}}$

Ans. D

Sol. $\log_e y = 3 \sin^{-1} x$

$$y = e^{3 \sin^{-1} x}$$

$$\frac{dy}{dx} = e^{3 \sin^{-1} x} \cdot \frac{3}{\sqrt{1-x^2}}$$

$$\sqrt{1-x^2} \frac{dy}{dx} = 3y$$

Again differentiate

$$\sqrt{1-x^2} \cdot y'' - \frac{2x}{2\sqrt{1-x^2}} y' = 3y'$$

$$(1-x^2)^2 y'' - xy' = 3y'(\sqrt{1-x^2})$$

So value of $3y'(\sqrt{1-x^2})$

$$\text{at } x = \frac{1}{2}$$

$$3 \cdot \frac{3}{\sqrt{1-x^2}} e^{\sin^{-1} x} (\sqrt{1-x^2})$$

$$= 9e^{\frac{3\pi}{6}} = 9e^{\frac{\pi}{2}}$$

138. Suppose $f(x) = \frac{(2^x + 2^{-x}) \tan x \sqrt{\tan^{-1}(x^2 - x + 1)}}{(7x^2 + 3x + 1)^3}$. Then the value of $f'(0)$ is equal to :

माना $f(x) = \frac{(2^x + 2^{-x}) \tan x \sqrt{\tan^{-1}(x^2 - x + 1)}}{(7x^2 + 3x + 1)^3}$ है। तो $f'(0)$ का मान बराबर है :

- (A) $\frac{\pi}{2}$ (B) π (C) 0 (D) $\sqrt{\pi}$

Ans. D

Sol. Let $f(x) = \tan x g(x)$

$$\text{where } g(x) = \frac{(2^x + 2^{-x})\sqrt{\tan^{-1}(x^2 - x + 1)}}{(7x^2 + 3x + 1)^3}$$

$$\Rightarrow f'(x) = \sec^2 x g(x) + \tan x g'(x)$$

$$f'(0) = \sec^2(0)g(0) + \tan 0 g'(0)$$

$$= g(0)$$

$$g(0) = \frac{(2^0 + 2^{-0})\sqrt{\tan^{-1}(0^2 - 0 + 1)}}{(7(0)^2 + 3(0) + 1)^3}$$

$$= 2\sqrt{\tan^{-1}1} = 2\sqrt{\frac{\pi}{4}} = \sqrt{\pi}$$

139. Let $y = \log_e \left(\frac{1-x^2}{1+x^2} \right)$, $-1 < x < 1$. Then at $x = \frac{1}{2}$, the value of $225(y' - y'')$ is equal to :

माना $y = \log_e \left(\frac{1-x^2}{1+x^2} \right)$, $-1 < x < 1$ है। तो $x = \frac{1}{2}$ पर $225(y' - y'')$ का मान है :

- (A) 736 (B) 746 (C) 732 (D) 742

Ans. A

Sol. $y = \log_e \left(\frac{1-x^2}{1+x^2} \right)$

$$\frac{dy}{dx} = y' = \frac{-4x}{1-x^4}$$

$$\text{Again } \frac{d^2y}{dx^2} = y'' = \frac{-4(1+3x^4)}{(1-x^4)^2}$$

$$\text{Again } y' - y'' = \frac{-4x}{1-x^4} + \frac{4(1+3x^4)}{(1-x^4)^2}$$

$$\text{at } x = \frac{1}{2}$$

$$y' - y'' = \frac{736}{225}$$

$$\text{Thus } 225(y' - y'') = 225 \times \frac{736}{225} = 736$$

140. If $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3 + \sin^2 2x \\ 3 + 2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3 + 2\sin^4 x & \sin^2 2x \end{vmatrix}$ then $\frac{1}{5}f'(0)$ is equal to :

यदि $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3 + \sin^2 2x \\ 3 + 2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3 + 2\sin^4 x & \sin^2 2x \end{vmatrix}$ है, तो $\frac{1}{5}f'(0)$ बराबर है :

(A) 2

(B) 6

(C) 0

(D) 1

Ans. C

Sol. $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3 + \sin^2 2x \\ 3 + 2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3 + 2\sin^4 x & \sin^2 2x \end{vmatrix}$

$$f'(x) = \begin{vmatrix} 8\cos^3 x (-\sin x) & 2\sin^4 x & 3 + \sin^2 2x \\ 8\cos^3 x (-\sin x) & 2\sin^4 x & \sin^2 2x \\ 8\cos^3 x (-\sin x) & 3 + 2\sin^4 x & \sin^2 2x \end{vmatrix} + \begin{vmatrix} 2\cos^4 x & 8\sin^3 x \cos x & 3 + \sin^2 2x \\ 3 + 2\cos^4 x & 8\sin^3 x \cos x & \sin^2 2x \\ 2\cos^4 x & 8\sin^3 x \cos x & \sin^2 2x \end{vmatrix} +$$

$$\begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 4\sin 2x \cos 2x \\ 3 + 2\cos^4 x & 2\sin^4 x & 4\sin 2x \cos 2x \\ 2\cos^4 x & 3 + 2\sin^4 x & 4\sin 2x \cos 2x \end{vmatrix}$$

$$f'(0) = \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 2 & 0 & 0 \\ 5 & 0 & 0 \\ 2 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 2 & 0 & 0 \\ 5 & 0 & 0 \\ 2 & 0 & 0 \end{vmatrix}$$

$$f(0) = 0$$

$$\therefore \frac{1}{5}f'(0) = 0$$